From Figure 4.3 we estimate that E_B/A for neon is about 8.0 MeV and for calcium about 8.6 MeV. Therefore, in the 2 neon nuclei $2 \times 20 \times 8.0 = 320$ MeV are involved in the binding energy, while $40 \times 8.6 = 344$ MeV binding energy are involved in the calcium nucleus. When 2 neon nuclei react to form the calcium nucleus the difference in the total binding energy of reactants and products is released; the estimate gives $344-320 = 24$ MeV; a calculation using measured masses gives 20.75 MeV. It is also evident that the greater gain in fusion energy is in the combination of lighter elements where the derivative of the E_B/A curve is greater, e.g. $^{2}H + {}^{2}H \rightarrow {}^{4}He$ liberates 23.85 MeV per atom of ^{4}He produced.

Figure 4.3 also shows that a similar release of binding energy can be obtained if the elements with mass numbers greater than 60 are split into lighter nuclides with higher E_B/A values. Such a process, whereby a nucleus is split into two smaller nuclides, is known as *fission*. An example of such a fission process is the reaction

$$^{236}_{92}U \rightarrow {}^{140}_{54}Xe + {}^{93}_{38}Sr + 3n \tag{4.12}$$

The binding energy per nucleon for the uranium nucleus is 7.6 MeV, while those for the ^{140}Xe and ^{93}Sr are 8.4 and 8.7 MeV respectively. The amount of energy released in this fission reaction is approximately $140 \times 8.4 + 93 \times 8.7 - 236 \times 7.6 = 191.5$ MeV for each uranium fission.

4.5. NUCLEAR RADIUS

Rutherford showed by his scattering experiments that the nucleus occupies a very small portion of the total volume of the atom. Roughly, the radii of nuclei vary from 1/10 000 to 1/100 000 of the radii of atoms. While atomic sizes are of the order of 100 pm (10^{-10} m), the common unit of nuclear size is the femtometer (1 fm = 10^{-15} m), sometimes referred to as 1 Fermi.

Experiments designed to study the size of nuclei indicate that the volumes of nuclei (V_n) are directly proportional to the total number of nucleons present, i.e.

$$V_n \propto A \tag{4.13}$$

Since for a sphere $V \propto r^3$, where r is the radius of the sphere, for a spherical nucleus $r^3 \propto A$, or $r \propto A^{1/3}$. Using r_o as the proportionality constant

$$r = r_o A^{1/3} \tag{4.14}$$

The implications of this is that the nucleus is composed of nucleons packed closely together with a constant density (about 0.2 nucleons fm^{-3}) from the center to the edge of the nucleus. This constant density model of the nucleus as been shown to be not completely correct, however. By bombarding nuclei with very high energy electrons or protons (up to $\geq$1 GeV) and measuring the scattering angle and particle energy, the

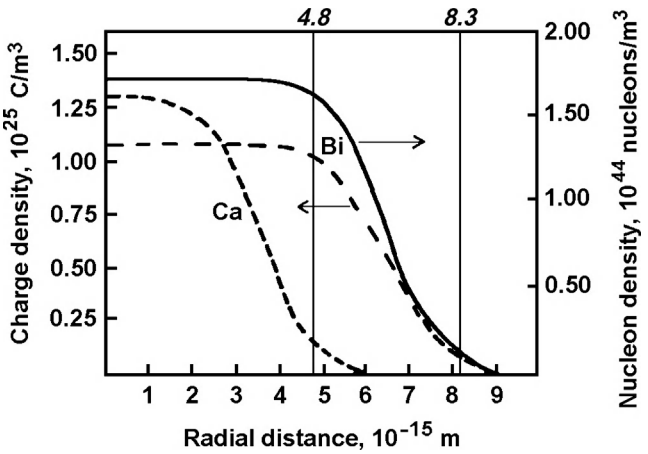


Figure 4.4 Experimentally measured charge and nuclear density values for ^{40}Ca and ^{209}Bi as a function of the nuclear radius.

charge and matter density near the surface of the irradiated nucleus can be studied. These experiments have led to the conclusion that nuclei do not possess a uniform charge or matter distribution out to a sharp boundary, but rather are fuzzy as indicated by the s-shaped curves in Figure 4.4. With an atomic number greater than 20 it has been found that a uniform charge and mass density exists over a short distance from the center of the nucleus, and this core is surrounded by a layer of decreasing density which seems to have a constant thickness of ~2.5 fm independent of mass number. In a bismuth nucleus, for example, the density remains relatively constant for approximately 5 fm then decreases steadily to one-tenth of that value in the next 2 fm (Fig. 4.4). This is not too surprising considering that as we get closer and closer to the edges of the nucleus not all the coordination sites of the nucleons are filled and thus the density will slowly decrease. It has also been found that not all nuclei are spherical, some being oblate and others prolate around the axis of rotation.

Despite the presence of this outer layer of decreasing density and the nonspherical symmetry, for most purposes it is adequate to assume a constant density nucleus with a sharp boundary. Therefore, use is made of the radius equation (4.14) in which the r_o value may be assumed to be 1.4 fm. Using this relationship, we can calculate the radius of ^{40}Ca to be $r = 1.4 \times 10^{-15} \times 40^{1/3} = 4.79$ fm, and for ^{209}Bi to be 8.31 fm. These values are indicated in Figure 4.4. For ^{80}Br a similar calculation yields 6.0 fm, while for ^{238}U the radius calculated is 8.7 fm. From these calculations we see that the radius does not change dramatically from relatively light nuclei to the heaviest.

4.6. SEMIEMPIRICAL MASS EQUATION

In preceding sections we have learned that the size as well as the total binding energy of nuclei are proportional to the mass number. These characteristics suggest an analogy